Ex. no. 4

Practical - 4

AIM:

Setup and configure a LAN (Local Area Network) using a Switch and Ethernet cables in your lab.

What is a LAN?

A local Area network (LAN) refers to a network that connects devices within a limited area, such as an office building, school, or home. It enables users to share resources, including data, printers and internet access. LAN connects devices to printers, servers and switches. A local area network (LAN) switch serves as the primary connecting device, managing and directing communications within the local network. Each connected device on a LAN switch can communicate directly with each other, allowing for fast and secure data transfer.

How to set up a LAN

Step 1: Plan and design an appropriate network topology taking into account network requirements and equipment location

Step 2: You can take 4 computers, a Switch with 8, 16 or 24 ports which is sufficient for network of these sizes and 4 Ethernet cables

Step 3: Connect your computers to network switch via Ethernet cable, which is as saimple as plugging one end of the ethernet cable into your computer and

the other end into your network switch

Step 4: Assign IP address to your PCs

1. Log on to the client computer as Administrator or as owner.
2. Click Network and Internet connections
3. Right click Local Area Connection/Ethernet → Go to Properties → select Internet Protocol (TCP/IPV4) → Click on properties → Select use the following IP address option and assign ipaddress

Step 5: Configure a network switch

1. Connect your computer to the switch: To access the switch using an Ethernet cable
2. Log in to the web interface: Open a web browser and enter the IP address of the switch in the address bar this should bring up the login page for the switch's web interface. Enter the username and password to log in
3. Configure basic settings: Once you're loggind in, you will be able to configure basic settings for the switch
4. Assign IP address as 10.1.1.5, subnet mask 255.0.0.0

Step 6: Check the connectivity btwn switch and other machine by using ping command in the command prompt of the device.

Step 7: Select a folder → go to properties → Click Sharing tab → Share it with everyone on the same LAN

Step 8: Try to access the shared folder from other computers of the network.

STUDENT OBSERVATION

1. Draw a neat diagram of the LAN in the configuration observation book that you have implemented in your lab. write the IP configuration of each and every device. write the outcome and challenges faced while configuring the LAN

The LAN was implemented by connecting multiple by connecting multiple PCs to a central switch using Ethernet cables. Each PC was assigned a unique IP address in the same network

IP configuration

Device	IP address	Subnet Mask
PC1		
PC2		
PC3		

Outcome: The LAN was successfully configured, and the PCs were able to communicate with each other

Challenges faced: the main challenges were assigning unique IP addresses, using the correct subnet mask, and checking the physical cable connections